UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Fall 2023

Paper: Computational Physics-II
Course Code: PHYS-3504 **Time: 3 H**

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PR

Q.1. Write short answers of the following Questions in MATLAB.

i) To generate an array R. Find out zero & nonzero elements and index numbers of R. Find maximum value and index number of R.
ii) Write example to show the function of the following.
(a) min() (b) zeros() (c) size() (d) roots() and (e) diff()
iii) Write syntax with example for the following in MATLAB:
(a) for() loop (b) ginput() and (c) gtext()
iv) Create a vector H with values 100 to 1 having decrement 2. Find sum of H, plot H and print H in reverse order.
v) To calculate by a function the area and circumference of a circle by taking the values of radius from the user
vi) Write one example to show the use of: poly(), polyfit() & rand()

Answer the following questions. (3x10=30)

Q.2.	How randomly generated points can be used to show Brownian motion? Write MATLAB program to simulate Brownian motion of an object for 31 collisions. Plot estimate graph. Write a program to find out factor of a number. Write MATLAB program to evaluate $\int_1^6 (3\sqrt{x} - 0.5)dx$.
Q.3.	Suppose A be a 3x3 matrix. Write MATLAB program which reads in random numbers as entries of the matrix A and calculate (i) sum and average of the all matrix elements, (ii) transpose of the matrix A (also plot the matrix), (iii) also check whether the Matrix A is an identity matrix? (iv) sort the matrix elements, (v) divide matrix rows by its row average. Write program to study the damped harmonic motion (DHM) of a mass attached with a spring using Euler's method with: (g=9.8 m/s^2, initial position zero and velocity 15 m/s, time step 0.1 sec. and tmax 15 sec., k = 1 N/m, m=1kg, damping coefficient = 0.5 N/ms,). Print and plot values for time, position, velocity and acceleration. Also suggest changes to write program for Simple harmonic motion.
Q.4.	Write MATLAB program to plot and print time, position, velocity and acceleration values for a spherical object in air drag. The acceleration and other parameters are given by $a = g - k v^2$ where $k = c \pi \rho r^2 / 2 m$ with the conditions: g = 9.8 m/sec^2, c=0.46 (drag constant), ρ= 1.2 kg/m^3, r=1m, v=0 m/sec, h= 0.1 sec and tmax=2.5sec. Write code to solve for the system of equations for x, y and z by using two different methods.Such that 2x+y+2z =17, 4x-3y+8z=6 and x-3y+z=6.

BS4Y

Computational Physics - II

(Fall - 2023)

Short Questions :-

(i)

```
>> % Generate an array R
R = [0 3 0 7 2 0 9 4];
zero_elements = R(R == 0);
zero_indices = find(R == 0);
nonzero_elements = R(R ~= 0);
nonzero_indices = find(R ~= 0);
[max_value, max_inden] = max(R);
disp('Zero elements and their
      indices:');
disp(zero_elements);
disp(zero_indices);
disp('Non zero elements and their
      indices:');
disp(nonzero_elements);
disp(non zero_indices);
disp('maximum value and its
      inden:');
disp(max_value);
disp(max_inden);
```

(ii)

```
>> % Define vector A
   A = [3 5 2 8 4];
   minValue = min(A)
   zeroMatrix = zeros(2,3)
   B = [1 2 3; 4 5 6];
   SizeB = size(B)
   C = [1 -3 2];
   PolinomialRoots = roots(C)
   D = [10 20 30 40];
   differenceD = diff(D)
```

(iii)

Syntax:-

```
for index = start_value : end_value
    % statement
end
```

(a)

(b) `[x,y] = ginput(n)`

(c) `gtext('text')`

Example :-

```
>> % create a blank figure
        window
   figure;
   for i = 1:3
       [x,y] = ginput(1);
       plot(x,y,'xo');
       gtext(['Point', num2str(i)]);
   end
```

(iv)

```
>> % Create vector H with values
        from 100 to -1
        decrementing by 2
H = 100:-2:-1;
Sum_H = sum(H);
figure;
Plot(H)
title('Plot of H');
xlabel('index');
ylabel('value');
H_reverse = flip(H);
disp(H_reverse)
disp(Sum_H)
```

(v)

1. from the user to input radius

```
radius = input('Enter radius:');
A = pi * radius^2;
C = 2 * pi * radius;
disp(A);
disp(C);
```

(vi)

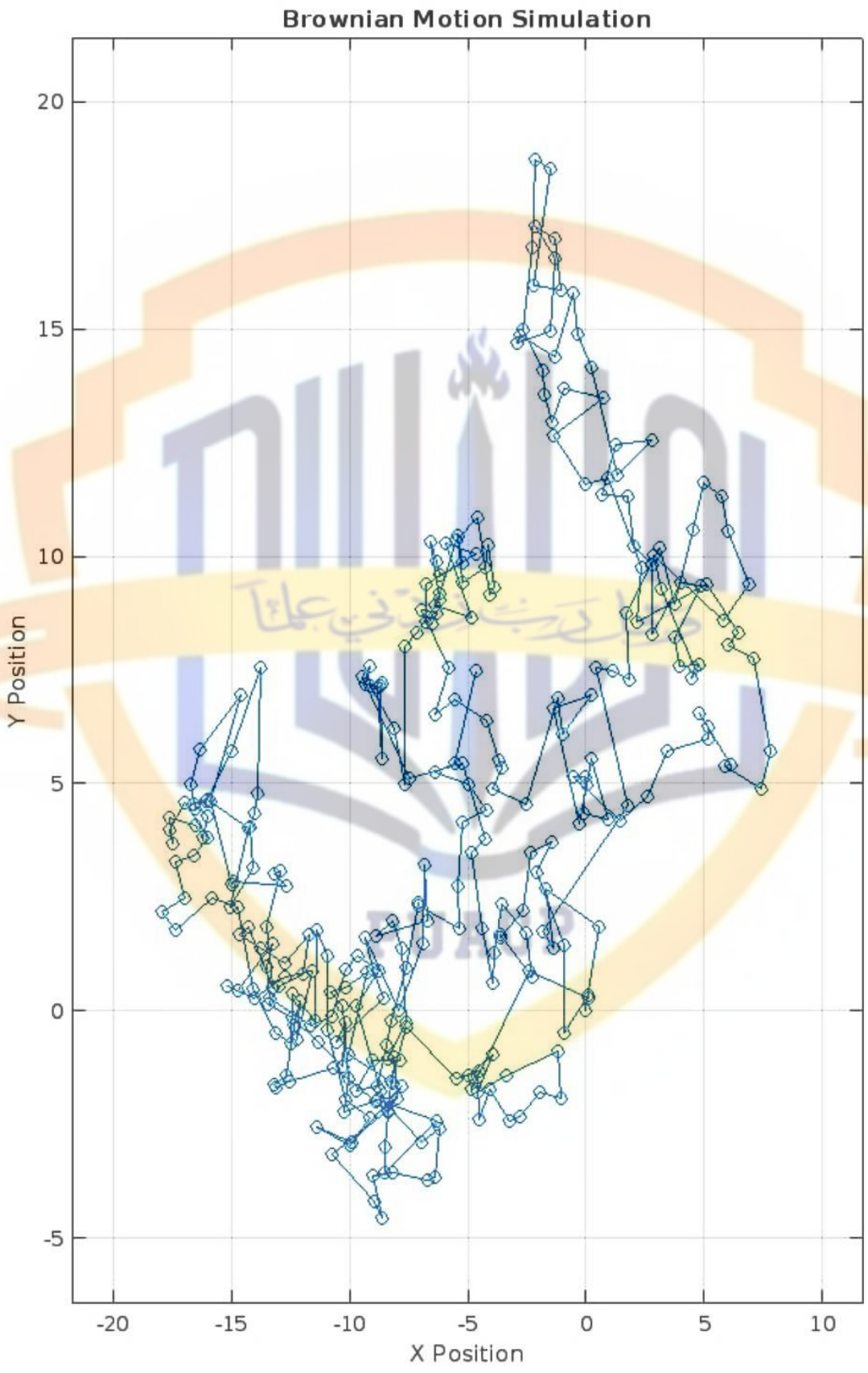
```
x = rand(1,5);
y = rand(1,5);
P = Polyfit(x, y, 2);
disp(P);
r = roots(P);
Polinomial = Poly(r);
disp(Polinomial);
x_fit = linspace(min(x), max(x),
                 100);
y_fit = polyval(P, x_fit)
figure;
plot(x, y);
hold on;
plot(x_fit, x_fit);
```

```
legend show;
title('Data and Fitted polynomial'
                                );
xlabel('x');
ylabel('y');
grid on;
```

1st long part (a)

```
>> n = 31 ;
   x = zeros(1, n);
   y = zeros(1, n);
   for i = 2:n
       daX = randn;
       daY = randn;
       x(i) = x(i-1) + daX;
       y(i) = y(i-1) + daY;
   end
   figure;
   plot(x, y, '-o');
   title('Brownian Motion
   Simulation');
   xlabel('X Position');
   ylabel('Y Position');
   grid on;
   figure;
   d = sqrt(x.^2 + y.^2);
   plot(1:n, d, '-o');
   title('Estimated Displacement
   from Origin');
   xlabel('Number of Steps');
   ylabel('Displacement');
   grid on
```

Output


Brownian Motion Simulation
20
15
10
5
0
-5
Y Position
-20
-15
-10
-5
0
5
10
X Position

1st long part (b)

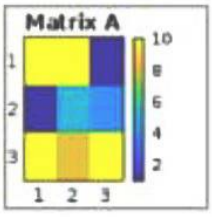
```
>> f = @(x) 3*sqrt(x) - 0.5;
   a = 1;
   b = 4;
   D= integral(f, a, b);
   disp(D)

    12.5000
```

2nd long part (a)

```
>> % 1. Generate a 3x3 matrix A with
   random numbers between 0 and 10
   A = randi([0, 10], 3, 3);
   disp(A);
   S= sum(A(:));
   avg = mean(A(:));
   disp(S)
   disp(avg);
   B= A';
   disp(B);
   figure;
   imagesc(A);
   colorbar;
   title('Matrix A')
   is_identity = isequal(A, eye(3));
   if is_identity
       disp('Matrix A is an identity
   matrix.');
   else
       disp('Matrix A is NOT an
   identity matrix.');
   end
   C=sort(A(:));
   disp(C);
   row_averages = mean(A, 2);
   A_divided = A ./ row_averages;
   disp(A_divided)
```

Output

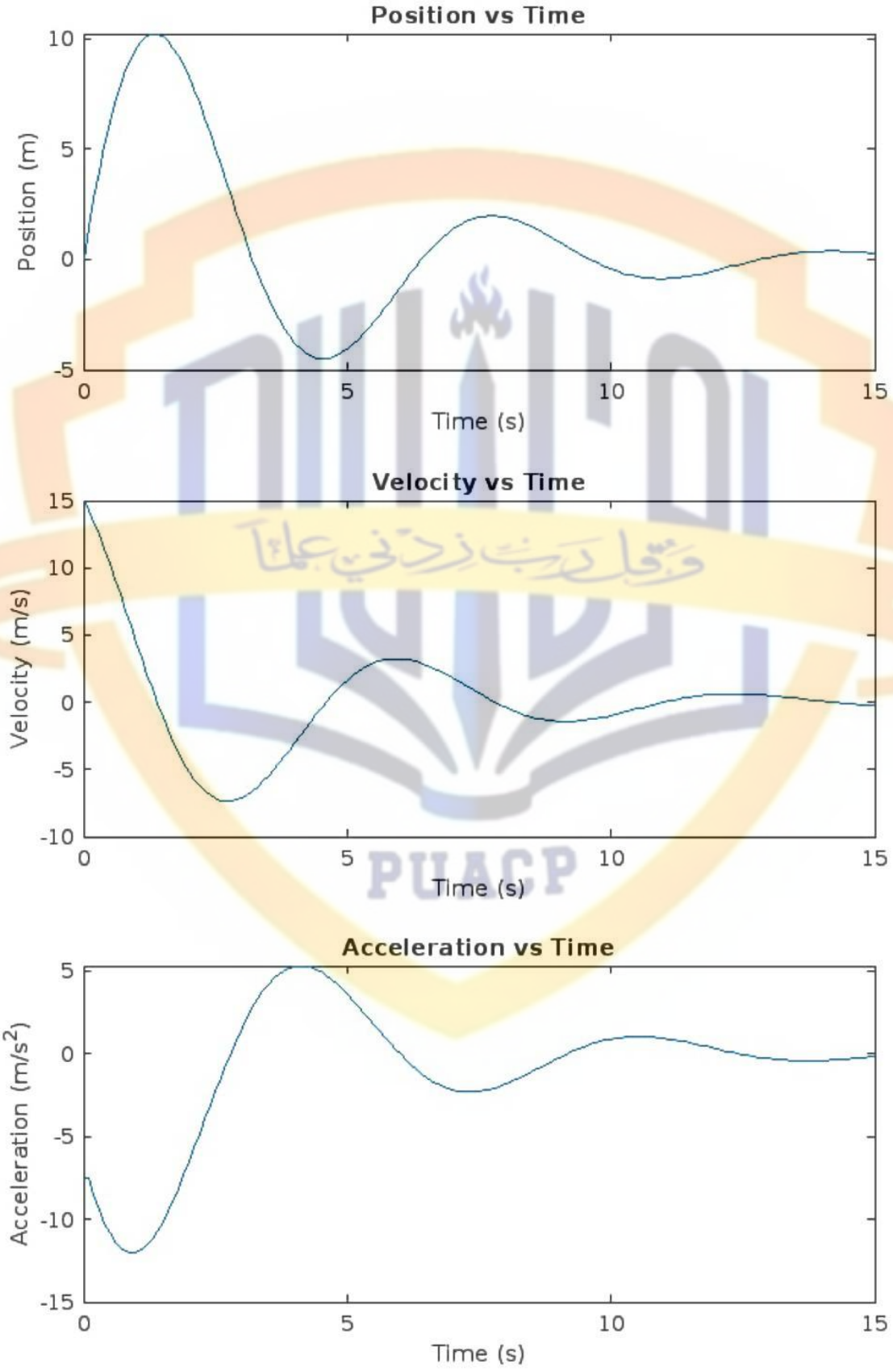
```
    10     10      1
     1      5      4
    10      8     10

    59

    6.5556

    10      1     10
    10      5      8
     1      4     10

Matrix A is NOT an identity matrix.
     1
     1
     4
     5
     8
    10
    10
    10
    10

    1.4286    1.4286    0.1429
    0.3000    1.5000    1.2000
    1.0714    0.8571    1.0714
```



2nd long part (b)

```
>> % DHM
    g = 9.8;
    k = 1;
    m = 1;
    c = 0.5;
    dt = 0.1;
    tmax = 15;
    x = 0;
    v = 15;
    time = 0:dt:tmax;
    n = length(time);
    x_vals = zeros(1, n);
    v_vals = zeros(1, n);
    a_vals = zeros(1, n);
    x_vals(1) = x;
    v_vals(1) = v;
    a_vals(1) = -(k/m)*x - (c/m)*v;
    for i = 2:n
        a = -(k/m)*x - (c/m)*v;
        v = v + a*dt;
        x = x + v*dt;
        x_vals(i) = x;
        v_vals(i) = v;
        a_vals(i) = a;
    end
    disp('Time (s)    Position (m)
    Velocity (m/s)    Acceleration (m/s^2)')
    for i = 1:n
        fprintf('%8.2f    %12.4f
    %13.4f    %18.4f\n', time(i), x_vals(i),
    v_vals(i), a_vals(i))
    end
    figure;
    subplot(3,1,1);
    plot(time, x_vals);
    xlabel('Time (s)');
    ylabel('Position (m)');
    title('Position vs Time');
    subplot(3,1,2);
    plot(time, v_vals);
    xlabel('Time (s)');
    ylabel('Velocity (m/s)');
    title('Velocity vs Time');
    subplot(3,1,3);
    plot(time, a_vals);
    xlabel('Time (s)');
    ylabel('Acceleration (m/s^2)');
    title('Acceleration vs Time')
```

Output


Position vs Time
Position (m)
Time (s)
Velocity vs Time
Velocity (m/s)
Time (s)
Acceleration vs Time
Acceleration (m/s^2)
Time (s)

3rd long part (a)

```
>> % under gravity and drag force
   g = 9.8;
   Cd = 0.46;
   rho = 1.2;
   r = 1;
   v0 = 0;
   h = 0.1;
   tmax = 2.5;
   A = pi * r^2;
   k = Cd * (rho * A) / (2 * 1);
   t = 0:h:tmax;
   N = length(t);
   v = zeros(1, N);
   x = zeros(1, N);
   a = zeros(1, N);
   v(1) = v0;
   x(1) = 0;
   for i = 2:N
       a(i-1) = g - k * v(i-1)^2;
       v(i) = v(i-1) + a(i-1) * h;
       x(i) = x(i-1) + v(i-1) * h;
   end
   a(N) = g - k * v(N)^2;
   figure;
   subplot(3,1,1);
   plot(t, x);
   xlabel('Time (s)');
   ylabel('Position (m)');
   title('Position vs Time');
   subplot(3,1,2);
   plot(t, v);
   xlabel('Time (s)');
   ylabel('Velocity (m/s)');
   title('Velocity vs Time');
   subplot(3,1,3);
   plot(t, a);
   xlabel('Time (s)');
   ylabel('Acceleration (m/s^2)');
   title('Acceleration vs Time');
   for i = 1:N
       fprintf( t(i), x(i), v(i), a(i));
   end
```

3rd long part (b)

```
>> % Cramer rule
   A = [2 1 2;
        4 -3 8;
        1 -3 1];
   det_A = det(A);
   Ax = A;
   Ax(:,1) = b;
   det_Ax = det(Ax);
   Ay = A;
   Ay(:,2) = b;
   det_Ay = det(Ay);
   Az = A;
   Az(:,3) = b;
   det_Az = det(Az);
   x= det_Ax / det_A;
   y= det_Ay / det_A;
   z= det_Az / det_A;
   disp('x,y,z')

       4 -3 8;
```

3rd long part (b)
second method

```
>> A = [2 1 2;
       4 -3 8;
       1 -3 1];
b = [17; -6; -6];

x = inv(A) * b;

disp(x);
```